

Wearables for various professions

Almost every profession has customised wearable technology solutions these days. Here are some examples.

Military. In the army, wearable communication tools, toxic gas sensors and cameras are quite common. A Harvard team has been recently funded to develop a smart suit for soldiers that can improve their body's resistance to injuries and delay the onset of fatigue. The proposed suit will comprise soft, wearable assistive devices that integrate several Wyss Institute technologies, including a stretchable sensor that monitors the body's biomechanics without the need for rigid, motion-restricting components. Recently, the US Army also developed smart under-clothes for soldiers, which use gel-free sensors to form an electronic network that monitors respiration, heart rate, body posture, and skin temperature. Computer algorithms are used to extrapolate this data, to assess fitness and impart better training. There has also been some news about the UK military developing solar-powered uniforms.

Surgeons. There have been quite a few examples of wearable technology that helps remote doctors guide mentee surgeons or even robots to perform tele-surgery. Recently, sensors-based tools have been developed to help doctors sense the pressure, temperature and other parameters of tissues more precisely, for more accurate surgery. One example is the electronic fingertip developed through a joint effort by the University of Illinois at Urbana-Champaign, Northwestern University, and China's Dalian University of Technology. The device consists of ultra-flexible silicon-based electronics and soft sensors mounted onto artificial skin that can be moulded to fit the wearer's fingers. It heightens the surgeon's sense of touch, and creates tingling sensations that trick the brain into feeling everything from pressure to temperature of the tissue being operated on. The same technology could also be used to develop gloves later.

Astronauts. Space-suits have always been heavily tech-laden and continue to be a hub of wearable tech innovation. Quite recently, NASA has funded a project called Electronic-textile System for the Evaluation of Wearable Technology (E-SEWT), which aims to bridge the gap between fashion and functionality in space suits. An astronaut donning an E-SEWT suit would be able to monitor anything on a spacecraft and interact directly with its systems. Critical controls and connections would be worn on the body where the astronaut can always access them. What's more, the suits will be comfortable and stylish.

Sportsmen. Sportsmen wear a lot of technology to help in their training. Suits that monitor speed and resilience, watches that keep track of training schedules, and goggles that keep tab of laps swum are common. Racers also wear GPS-enabled clothing, watches or shoes to track their location, speed, direction and more.

listening environments and to address specific tonal sensitivities.

In EFY's January issue, we had also discussed a new technology being developed by Cogni-Wave—a hearing-aid that uses rotating microphones designed to filter environmental sounds, so that the person using the hearing-aid hears exactly like someone with functional hearing, i.e., one voice at a time.

Four students from Ukraine have developed a simple and inexpensive solution called Enable Talk comprising special gloves and a mobile application, which can detect signs made by the user wearing the gloves, and convert it to text or speech on the phone. The glove is made up of eleven flex sensors and eight touch sensors, a 3D digital linear acceleration sensor, magnetic sensor, accelerometer and gyroscope in addition to an Atmel XMEGA A3 microcontroller, Bluetooth, built-in solar charger and other components.

On a more common plane, there are several wearable blood glucose meters from companies like Abbott and Medtronic, fitness and heart rate monitors from companies like Garmin, Adidas, Polar and Suunto, and customised vital-signs monitors, which are used either for personal monitoring or by doctors to monitor their wards. There are patches, head-gear, watches and a large range of health gear. Plus, the lifestyle monitors discussed earlier can also be used for various health-related purposes.

Wearable commerce. Last year also saw a couple of examples of how wearable technology could be used for commerce. Ticketholders to the Barclaycard Mercury Music Prize were given NFC wristbands known as 'Payband.' The aesthetically designed wristband enabled the wearer to gain entry to shows and buy stuff without carrying any cash or cards. The Payband is

basically an RFID (or NFC) microchip embedded in a silicone wristband, pre-loaded with payment-equivalent tokens. Similarly, Disney also recently introduced the MagicBand wristband that allows Disney World visitors to gain entry to rides or their hotel rooms, without any cash, cards or keys.

Motion capture. Garments and shoes with sensors and cameras can be used to capture the wearer's movements very closely. These are often used by sportspeople, gymnasts and dancers to critically review their work.

At CES this year, Xsens Technologies demonstrated a motion capture kit comprising a range of accelerometers, gyroscopes and magnetometer sensors plus powerful software. When a person wearing the sensors dances, a screen shows a vivid computer-animated figure that matches the moves. The idea is to offer users a way to digitise their actions and play them back as 3D graphics to help perfect their skills.

Culture first, tech next

It is evident from this discussion that there is no end to the fantastic options that wearable technology throws up. However, for such technologies to survive and thrive, technology makers have to understand that wearable technology is not merely a new technology or old technologies in a new package. It is a new plane of innovation, a new thought process, which puts man before machine.

As Sarah Rotman Epps, a senior analyst at Forrester, observes in her recent report *Smart Body, Smart World*: "Product strategists must embrace a human-centric approach to design—the person is the focus of innovation, not the device."

The wearable device should blend with your culture and lifestyle. For a wearable device to become successful, you should be able to wear it and use it without looking awkward. It should also serve a powerful purpose—because fashion is short-lived but functionality lasts. ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai